

Machine learning

Overfitting

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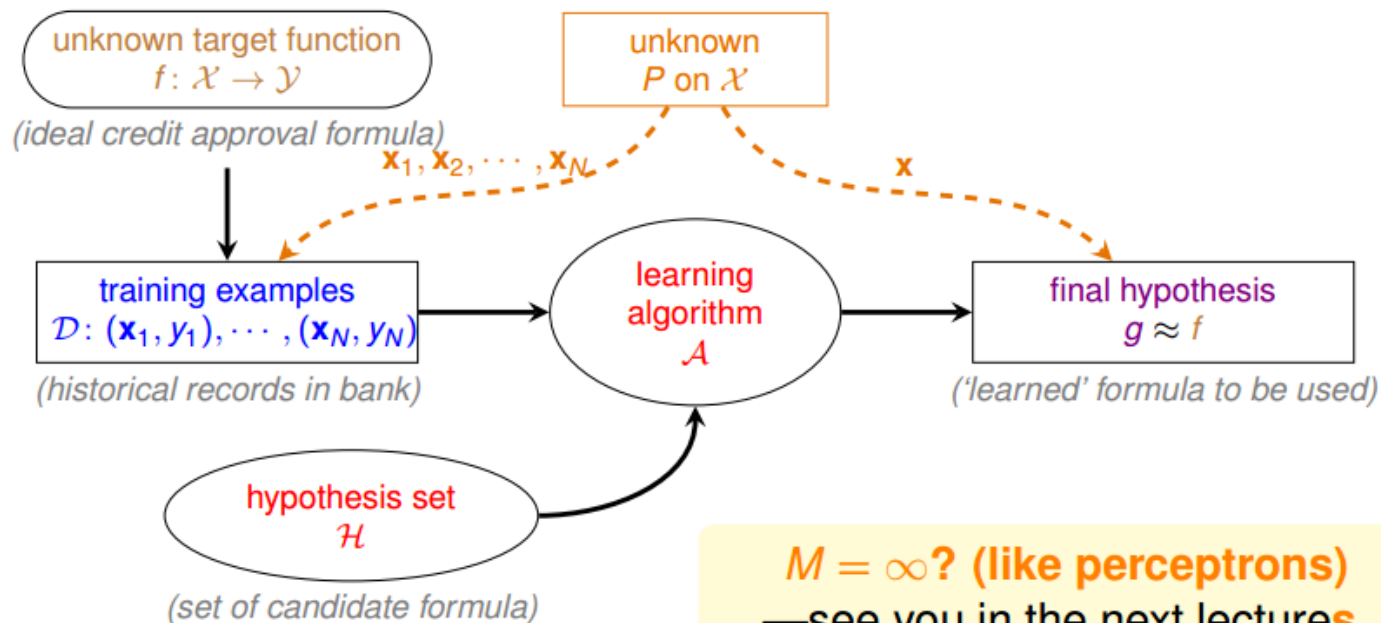


Contents

- Is learning feasible?

Learning?

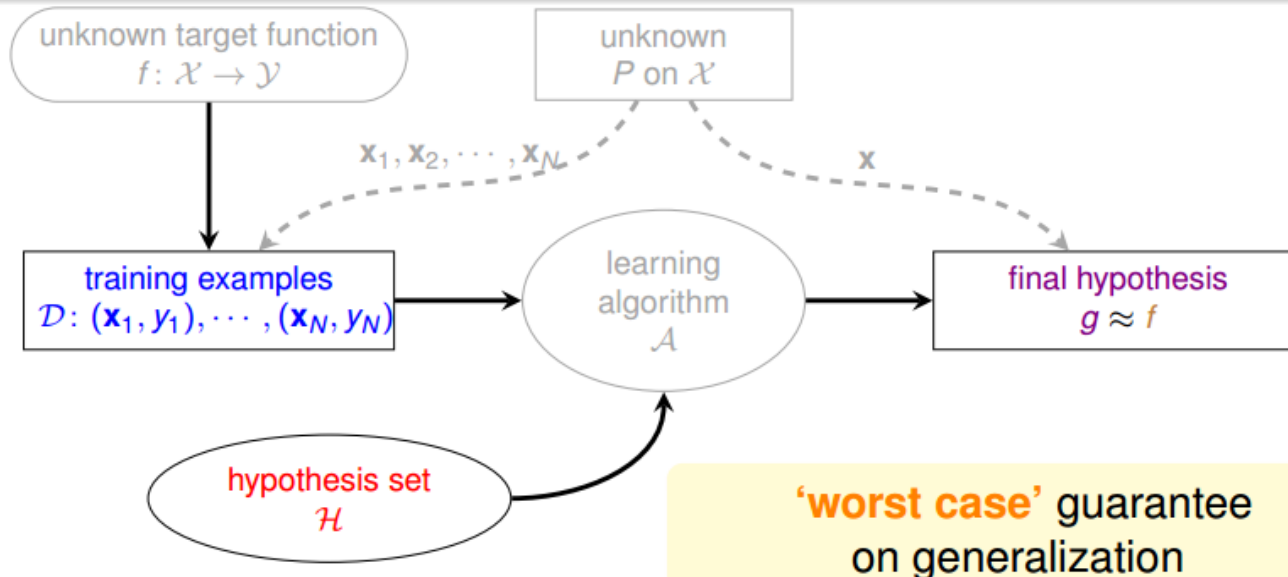
if $|\mathcal{H}| = M$ finite, N large enough,
for whatever g picked by $\mathcal{A}$, $E_{\text{out}}(g) \approx E_{\text{in}}(g)$
if $\mathcal{A}$ finds one g with $E_{\text{in}}(g) \approx 0$,
PAC guarantee for $E_{\text{out}}(g) \approx 0 \implies$ **learning possible :-)**



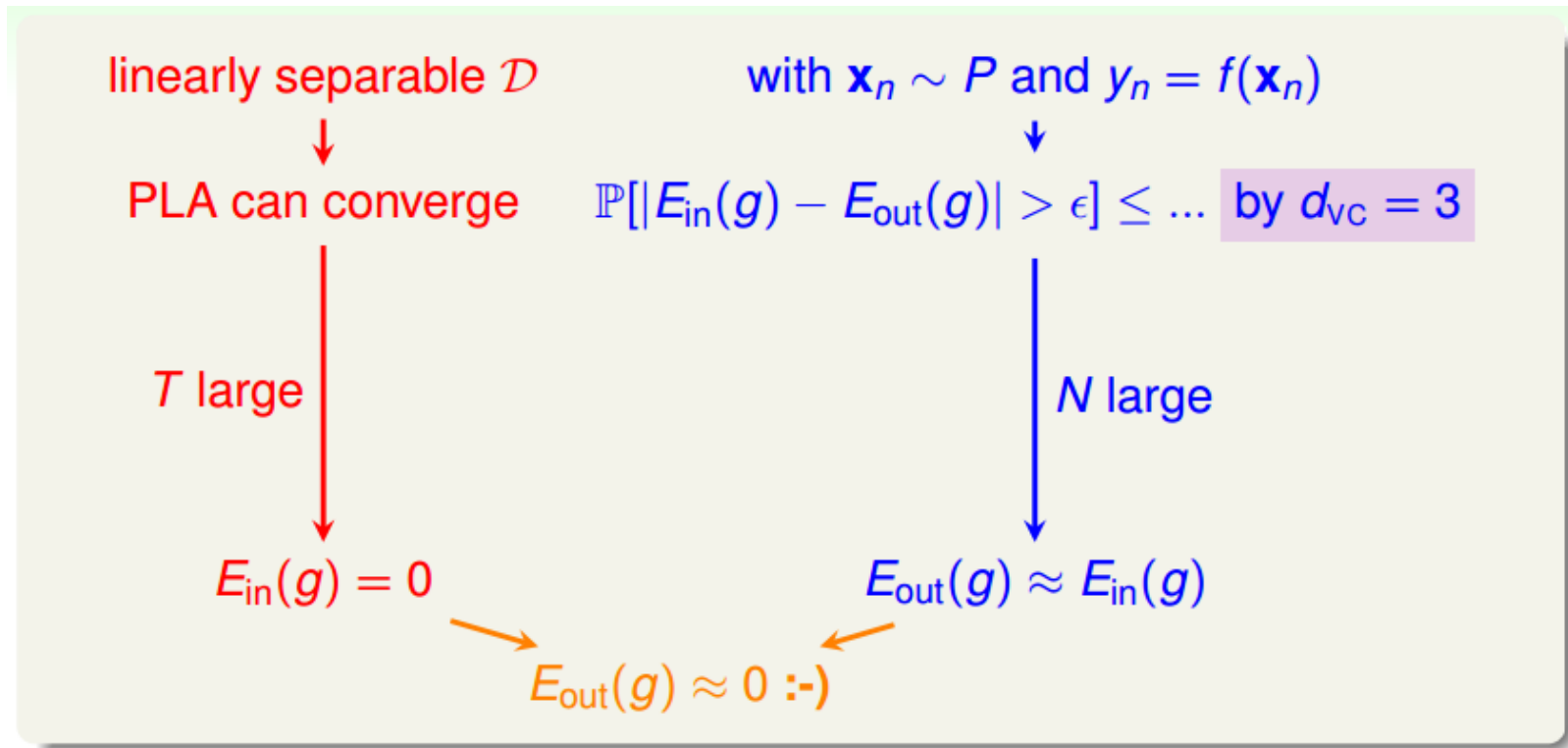
VC Dimension and Learning

finite $d_{VC} \implies g$ 'will' generalize ($E_{out}(g) \approx E_{in}(g)$)

- regardless of learning algorithm $\mathcal{A}$
- regardless of input distribution P
- regardless of target function f



2D PLA Revisited



general PLA for $\mathbf{x}$ with more than 2 features?

VC Dimension of Perceptrons

- 1D perceptron (pos/neg rays): $d_{VC} = 2$
- 2D perceptrons: $d_{VC} = 3$
 - $d_{VC} \geq 3$:

 - $d_{VC} \leq 3$:

- d -D perceptrons: $d_{VC} \stackrel{?}{=} d + 1$

two steps:

- $d_{VC} \geq d + 1$
- $d_{VC} \leq d + 1$

VC Bound Rephrase: Penalty for Model Complexity

For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2, d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)| > \epsilon}_{\text{BAD}} \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

Rephrase

..., with probability $\geq 1 - \delta$, **GOOD**: $|E_{in}(g) - E_{out}(g)| \leq \epsilon$

$$\text{set } \delta = 4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)$$

$$\frac{\delta}{4(2N)^{d_{VC}}} = \exp\left(-\frac{1}{8}\epsilon^2 N\right)$$

$$\ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right) = \frac{1}{8}\epsilon^2 N$$

$$\sqrt{\frac{8}{N} \ln\left(\frac{4(2N)^{d_{VC}}}{\delta}\right)} = \epsilon$$

VC Bound Rephrase: Penalty for Model Complexity

For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2, d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{\text{in}}(g) - E_{\text{out}}(g)|}_{\text{BAD}} > \epsilon \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

Rephrase

..., with probability $\geq 1 - \delta$, **GOOD!**

$$\text{gen. error } |E_{\text{in}}(g) - E_{\text{out}}(g)| \leq \sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{VC}}}{\delta} \right)}$$

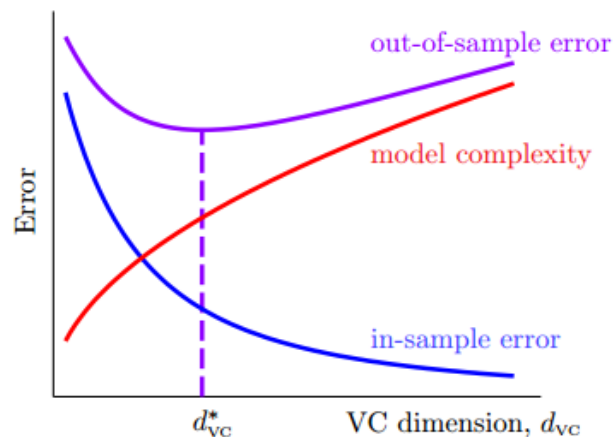
$$E_{\text{in}}(g) - \sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{VC}}}{\delta} \right)} \leq E_{\text{out}}(g) \leq E_{\text{in}}(g) + \sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{VC}}}{\delta} \right)}$$

$\underbrace{\sqrt{\dots}}_{\Omega(N, \mathcal{H}, \delta)}$: penalty for **model complexity**

THE VC Message

with a high probability,

$$E_{\text{out}}(g) \leq E_{\text{in}}(g) + \underbrace{\sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{\text{VC}}}}{\delta} \right)}}_{\Omega(N, \mathcal{H}, \delta)}$$



- $d_{\text{VC}} \uparrow$: $E_{\text{in}} \downarrow$ but $\Omega \uparrow$
- $d_{\text{VC}} \downarrow$: $\Omega \downarrow$ but $E_{\text{in}} \uparrow$
- best d_{VC}^* in the middle

powerful $\mathcal{H}$ not always good!

VC Bound Rephrase: Sample Complexity

For any $g = \mathcal{A}(\mathcal{D}) \in \mathcal{H}$ and 'statistical' large $\mathcal{D}$, for $N \geq 2$, $d_{VC} \geq 2$

$$\mathbb{P}_{\mathcal{D}} \left[\underbrace{|E_{in}(g) - E_{out}(g)|}_{\text{BAD}} > \epsilon \right] \leq \underbrace{4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right)}_{\delta}$$

given specs $\epsilon = 0.1$, $\delta = 0.1$, $d_{VC} = 3$, want $4(2N)^{d_{VC}} \exp\left(-\frac{1}{8}\epsilon^2 N\right) \leq \delta$

N	bound
100	2.82×10^7
1,000	9.17×10^9
10,000	1.19×10^8
100,000	1.65×10^{-38}
29,300	9.99×10^{-2}

sample complexity:
need $N \approx 10,000 d_{VC}$ in theory

practical rule of thumb:

$$N \approx 10 d_{VC} \text{ often enough!}$$

Looseness of VC Bound

$$\mathbb{P}_{\mathcal{D}} \left[|E_{\text{in}}(g) - E_{\text{out}}(g)| > \epsilon \right] \leq 4(2N)^{d_{\text{VC}}} \exp \left(-\frac{1}{8} \epsilon^2 N \right)$$

theory: $N \approx 10,000 d_{\text{VC}}$; practice: $N \approx 10 d_{\text{VC}}$

Why?

- Hoeffding for unknown E_{out} **any distribution, any target**
- $m_{\mathcal{H}}(N)$ instead of $|\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_N)|$ **'any' data**
- $N^{d_{\text{VC}}}$ instead of $m_{\mathcal{H}}(N)$ **'any' $\mathcal{H}$ of same d_{VC}**
- union bound on worst cases **any choice made by $\mathcal{A}$**

— **but hardly better, and 'similarly loose for all models'**

philosophical message of VC bound
important for improving ML

VC analysis is a **decomposition**.

Decompose E_{out} into E_{in} and ϵ .

$$E_{\text{out}} \leq E_{\text{in}} + \underbrace{\sqrt{\frac{8}{N} \log \frac{4((2N)^{d_{\text{VC}}} + 1)}{\delta}}}_{=\epsilon}$$

E_{in} = training error, ϵ = penalty of complex model.

Bias and variance is another decomposition.

Decompose E_{out} into

- How well can $\mathcal{H}$ approximate f ?
- How well can we zoom in a good h in $\mathcal{H}$?

Roughly speaking we will have

$$E_{\text{out}} = \text{bias} + \text{variance}$$

Bias and Variance

In **VC analysis** we define the out-sample error as

$$E_{\text{out}}(g) = \mathbb{E}_{\mathbf{x}} \left[\mathbf{1}\{g(\mathbf{x}) \neq f(\mathbf{x})\} \right]$$

Expectation of a **0-1 loss**.

In **Bias-variance** analysis we define the out-sample error as

$$E_{\text{out}}(g) = \mathbb{E}_{\mathbf{x}} \left[(g(\mathbf{x}) - f(\mathbf{x}))^2 \right].$$

Expectation of a **square loss**.

Square loss is differentiable.

Bias and Variance

The decomposition:

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{\text{out}}(g^{(\mathcal{D})}) \right] \\ = & \mathbb{E}_{\mathbf{x}} \left[\underbrace{\mathbb{E}_{\mathcal{D}} \left[g^{(\mathcal{D})}(\mathbf{x})^2 \right] - \bar{g}(\mathbf{x})^2}_{\mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]} + \underbrace{\bar{g}(\mathbf{x})^2 - 2\bar{g}(\mathbf{x})f(\mathbf{x}) + f(\mathbf{x})^2}_{(\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2} \right] \end{aligned}$$

Define two terms

$$\begin{aligned} \text{bias}(\mathbf{x}) &\stackrel{\text{def}}{=} (\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2, \\ \text{var}(\mathbf{x}) &\stackrel{\text{def}}{=} \mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]. \end{aligned}$$

Take expectation

$$\begin{aligned} \text{bias} &= \mathbb{E}_{\mathbf{x}}[\text{bias}(\mathbf{x})] = \mathbb{E}_{\mathbf{x}}[(\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2], \\ \text{var} &= \mathbb{E}_{\mathbf{x}}[\text{var}(\mathbf{x})] = \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]\right]. \end{aligned}$$



Bias and Variance Decomposition

The decomposition:

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{\text{out}}(g^{(\mathcal{D})}) \right] \\ = & \mathbb{E}_{\mathbf{x}} \left[\underbrace{\mathbb{E}_{\mathcal{D}} \left[g^{(\mathcal{D})}(\mathbf{x})^2 \right] - \bar{g}(\mathbf{x})^2}_{\mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]} + \underbrace{\bar{g}(\mathbf{x})^2 - 2\bar{g}(\mathbf{x})f(\mathbf{x}) + f(\mathbf{x})^2}_{(\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2} \right]. \end{aligned}$$

This gives

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{\text{out}}(g^{(\mathcal{D})}) \right] &= \mathbb{E}_{\mathbf{x}}[\text{bias}(\mathbf{x}) + \text{var}(\mathbf{x})] \\ &= \text{bias} + \text{var} \end{aligned}$$

Interpreting the Bias-Variance

The decomposition:

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{\text{out}}(g^{(\mathcal{D})}) \right] \\ = & \mathbb{E}_{\mathbf{x}} \left[\underbrace{\mathbb{E}_{\mathcal{D}} \left[g^{(\mathcal{D})}(\mathbf{x})^2 \right] - \bar{g}(\mathbf{x})^2}_{\mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]} + \underbrace{\bar{g}(\mathbf{x})^2 - 2\bar{g}(\mathbf{x})f(\mathbf{x}) + f(\mathbf{x})^2}_{(\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2} \right] \end{aligned}$$

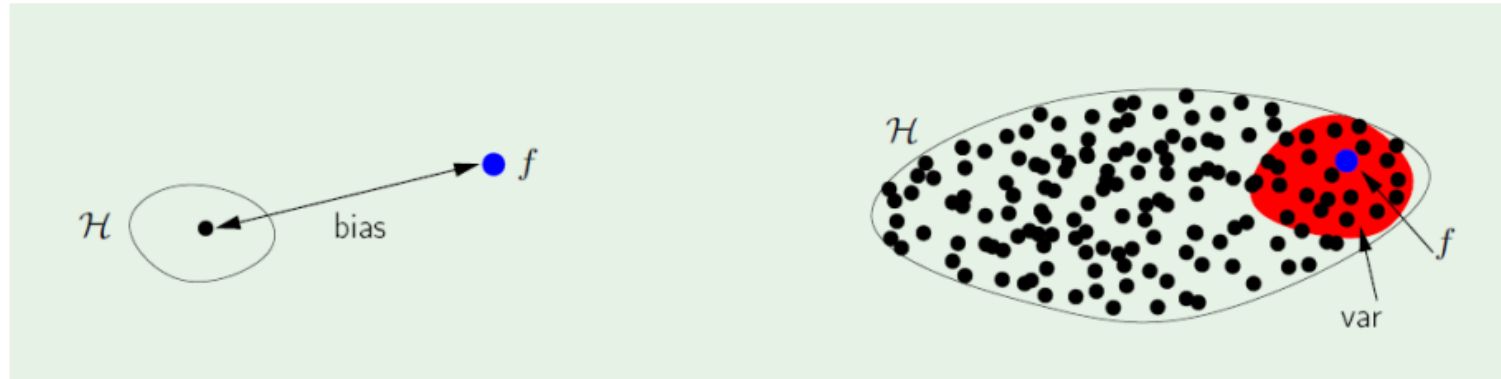
The two terms:

$$\begin{aligned} \text{bias}(\mathbf{x}) &\stackrel{\text{def}}{=} (\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2, \\ \text{var}(\mathbf{x}) &\stackrel{\text{def}}{=} \mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]. \end{aligned}$$

$\text{bias}(\mathbf{x})$: How close is the **average function** $\bar{g}$ to the target

$\text{var}(\mathbf{x})$: How much **uncertainty** you have around $\bar{g}$

Model Complexity



The bias and variance are

$$\text{bias}(\mathbf{x}) \stackrel{\text{def}}{=} (\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2,$$

$$\text{var}(\mathbf{x}) \stackrel{\text{def}}{=} \mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2].$$

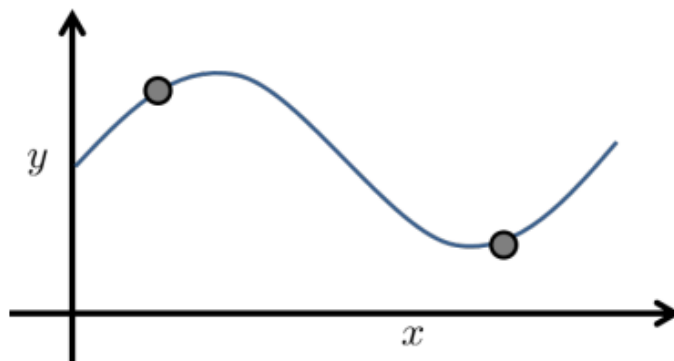
If you have a simple $\mathcal{H}$, then large bias but small variance

If you have a complex $\mathcal{H}$, then small bias but large variance

Example

Consider a $\sin(\cdot)$ function

$$f(x) = \sin(\pi x)$$



You are only given $N = 2$ training samples

These two samples are sampled uniformly in $[-1, 1]$.

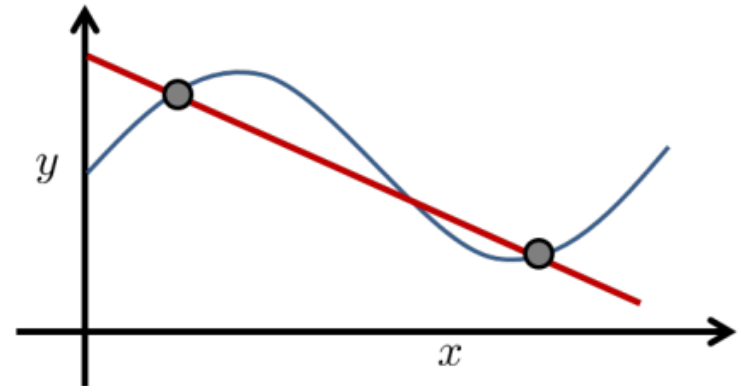
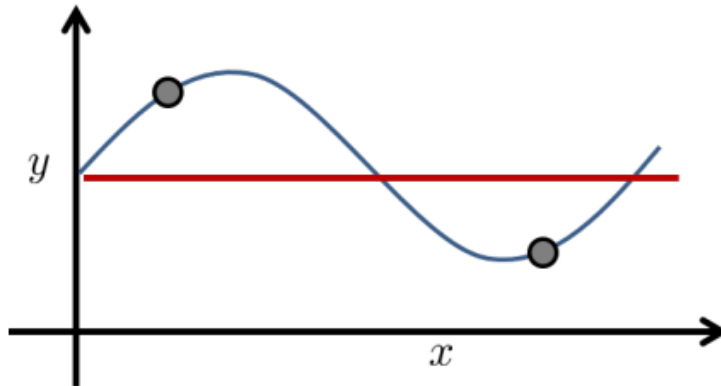
Call them (x_1, y_1) and (x_2, y_2)

Hypothesis Set 0: $\mathcal{M}_0$ = Set of all lines of the form $h(x) = b$;

Hypothesis Set 1: $\mathcal{M}_1$ = Set of all lines of the form $h(x) = ax + b$.

Which one fits better?

Example



If you give me two points, I can tell you the fitted lines

For $\mathcal{M}_0$:

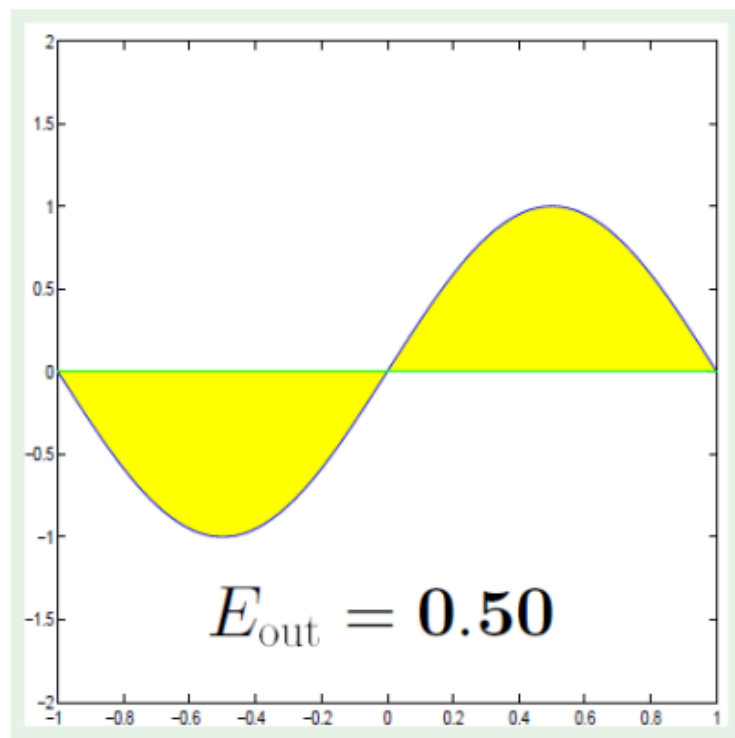
$$h(x) = \frac{y_1 + y_2}{2}.$$

For $\mathcal{M}_1$:

$$h(x) = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x + (y_1 x_2 - y_2 x_1).$$

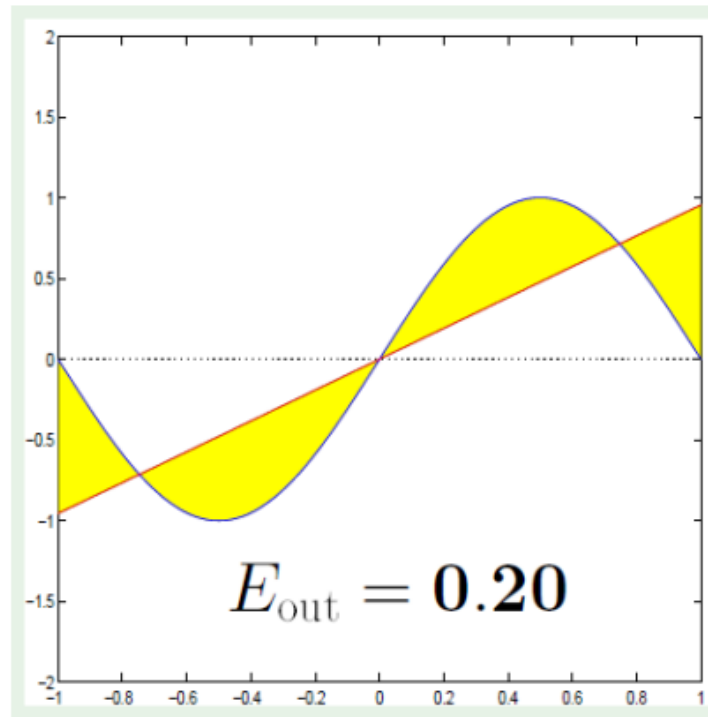
Out-sample Error E_{out}

If you use $\mathcal{M}_0$
Then you get this
 $E_{\text{out}} = 0.5$



Out-sample Error E_{out}

If you use $\mathcal{M}_1$
Then you get this
 $E_{\text{out}} = 0.2$



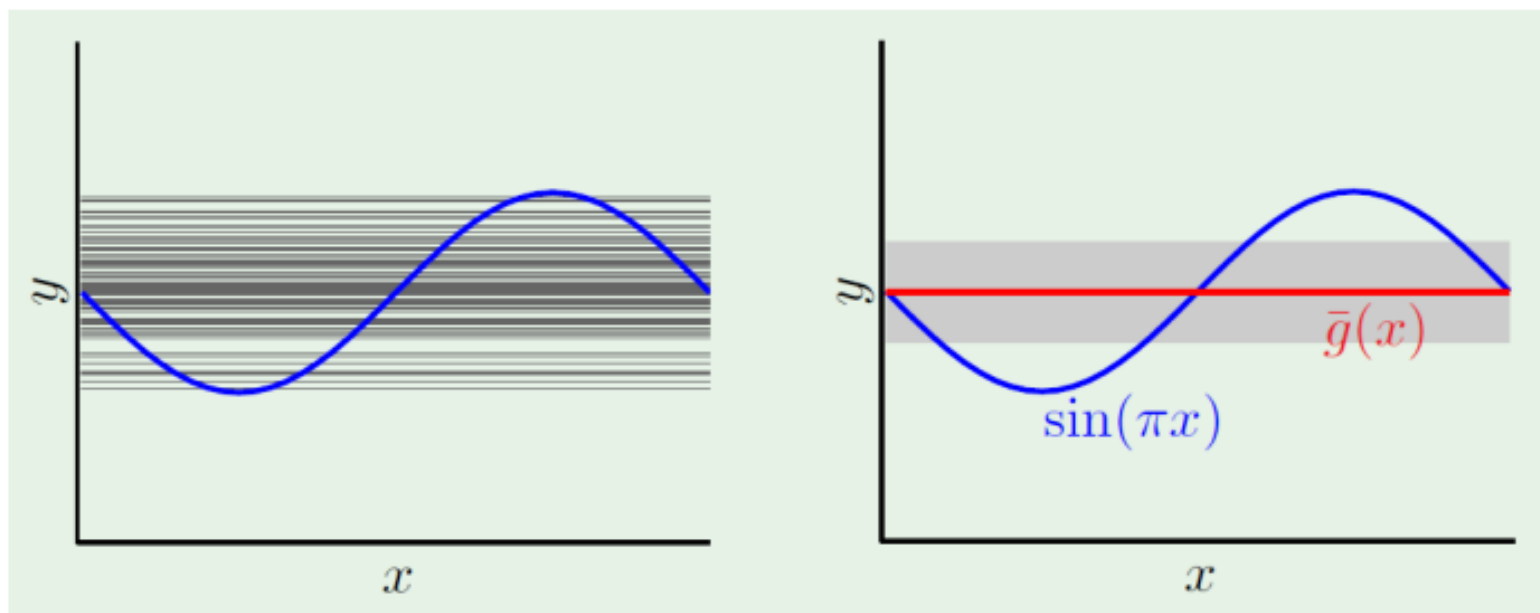
Scan through D

Now draw a different training set

Then you have a different curve every time

Plot them all on the same figure

Here is what you will get



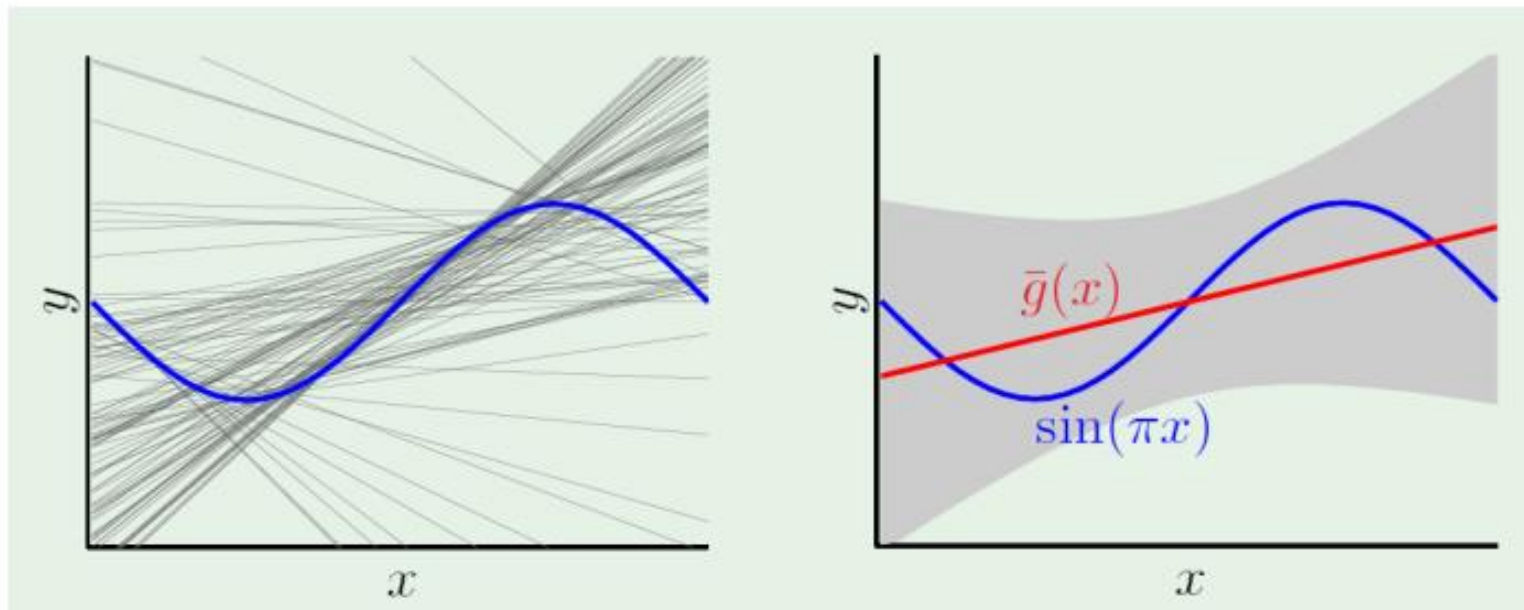
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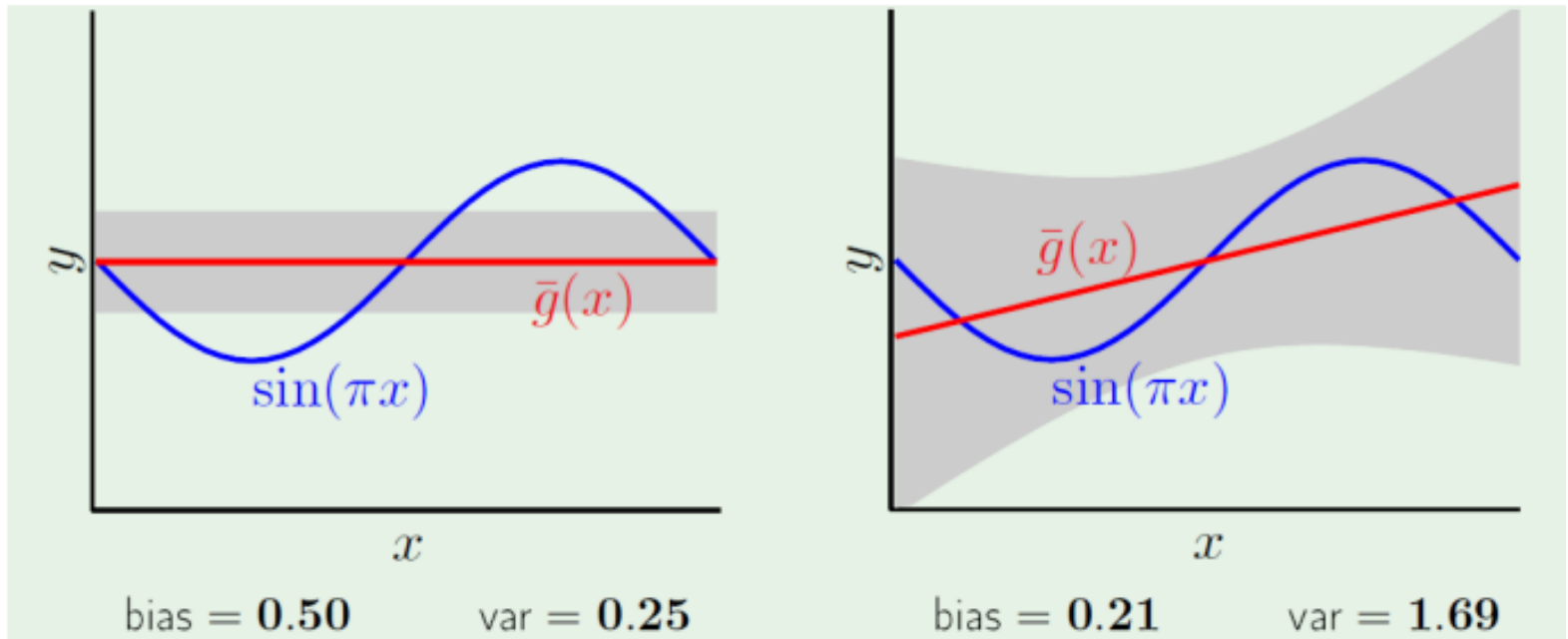
Limiting Case

Draw infinitely many training sets

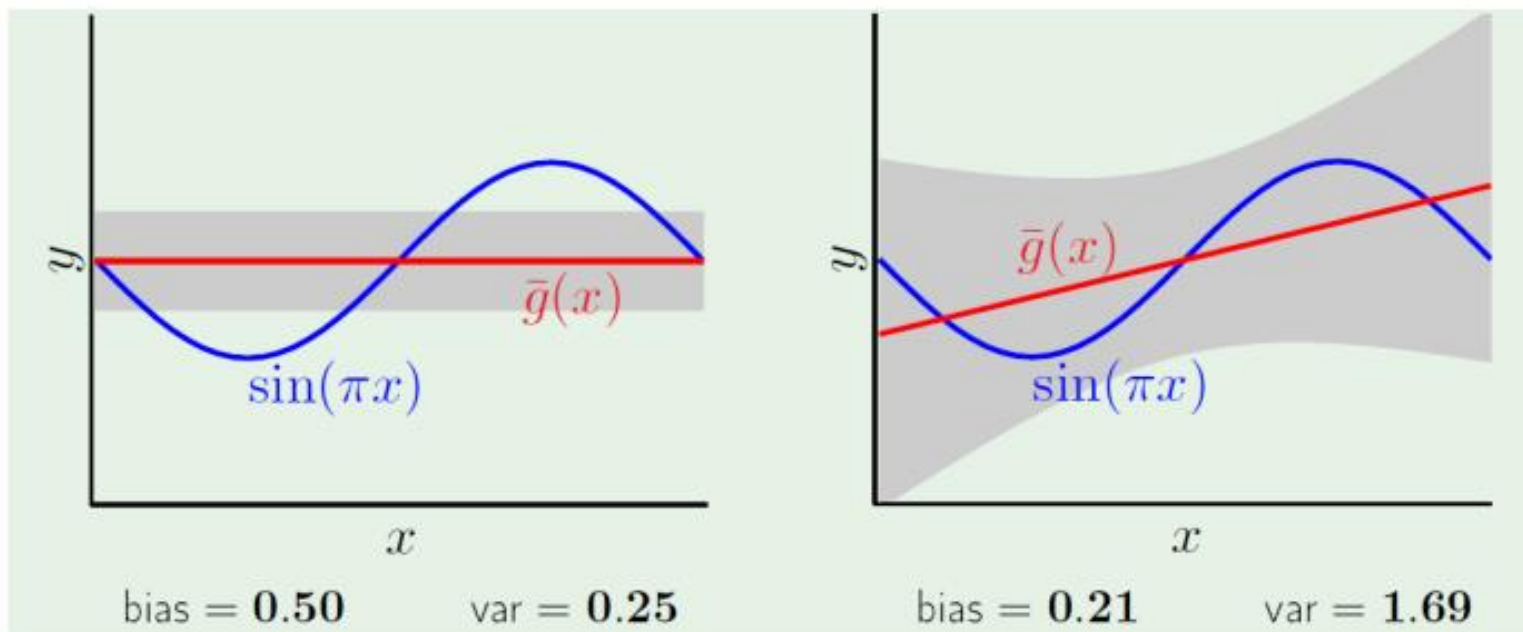
You will have two quantities

$\bar{g}(x)$: The average line

$\sqrt{\text{var}(x)}$: The variance



How Come!



$\bar{g}(x)$ is a good **average**.

But the **error bar** is big!

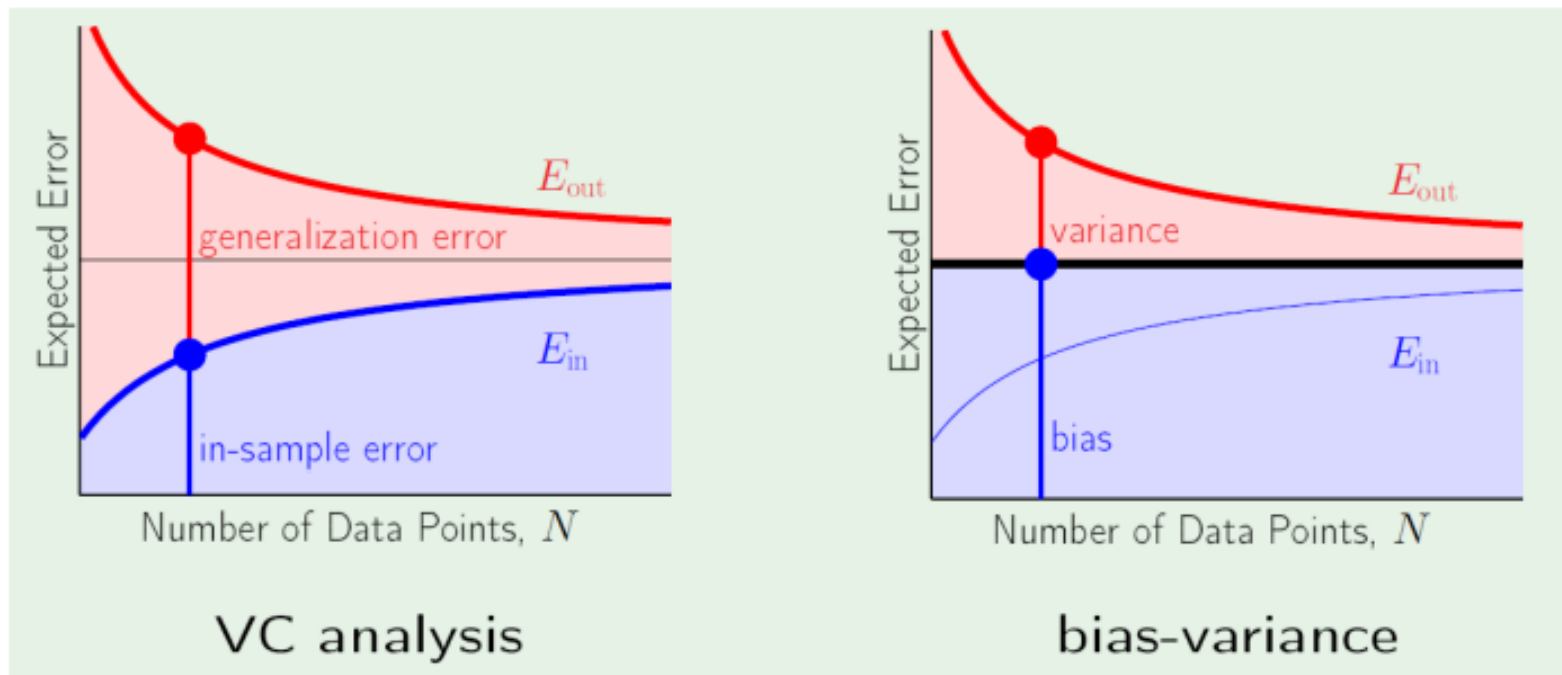
Analogy: I have a powerful canon but not very accurate.

Learning Curve

Expected out-sample error: $E_{\text{out}}(g^{(\mathcal{D})})$

Expected in-sample error: $E_{\text{in}}(g^{(\mathcal{D})})$

How do they change with N ?



VC vs Bias-Variance

- VC analysis is independent of $\mathcal{A}$
- Bias-variance depends on $\mathcal{A}$
- With the same $\mathcal{H}$, VC always returns the same generalization bound
- Guarantee over all possible choices of dataset $\mathcal{D}$
- Bias-variance: For the same $\mathcal{H}$, you can have different $g^{(\mathcal{D})}$
- Depend on $\mathcal{D}$, you have a different $E_{\text{out}}(g^{(\mathcal{D})})$
- Therefore we take expectation

$$\mathbb{E}_{\mathcal{D}} \left[E_{\text{out}}(g^{(\mathcal{D})}) \right]$$

- In practice, bias and variance cannot be computed
- You do not have f
- It is a conceptual tool to design algorithms

Overfit

- Source of Overfit

Is Noise the Reason?

Is Model Complexity the Reason?

The Trinity of Noise, Target Complexity, and Training Sample

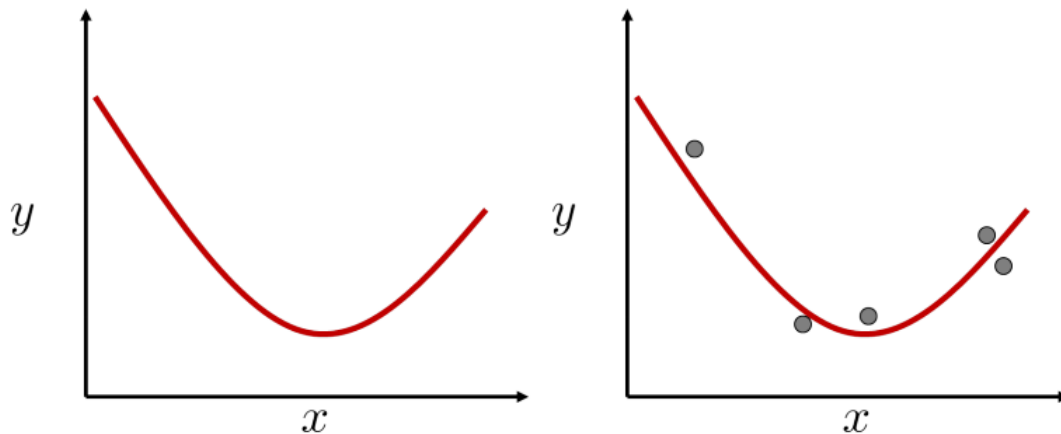
Analyzing Overfit

Bias and Variance

Learning Curve

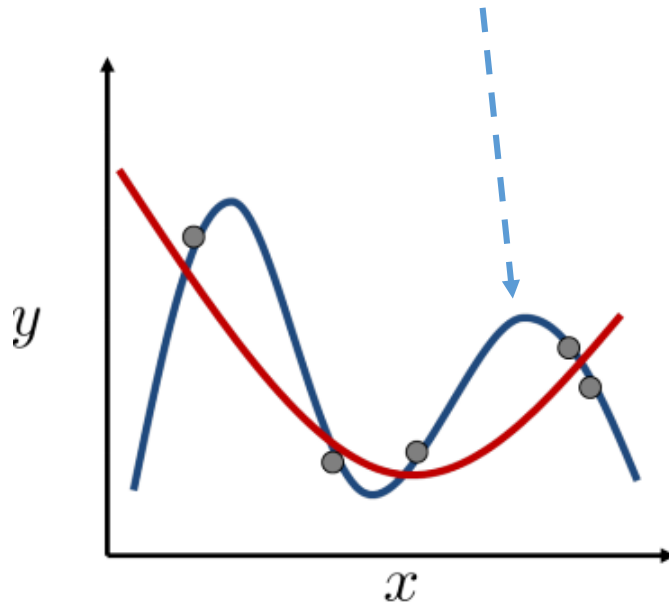
Case Study

- What is overfit?
- You have a simple target function f
- From this f you generate 5 data **noisy training samples**
- Then you use a 4-th order polynomial to fit the data

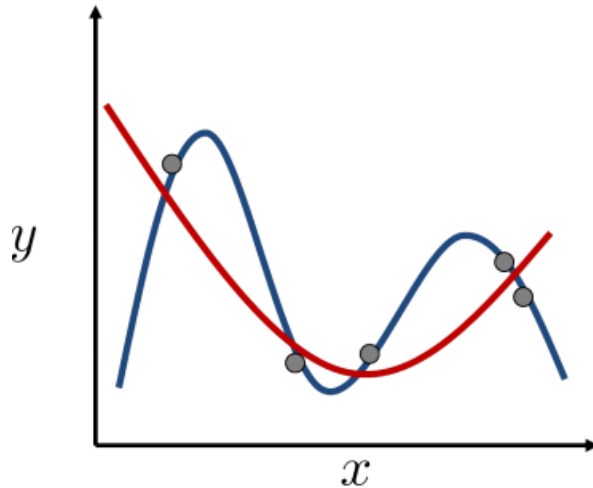


Case Study

- What is overfit?
- You have a simple target function f
- From this f you generate 5 data **noisy training samples**
- Then you use a **4-th order polynomial** to fit the data



What is Over-fitting?



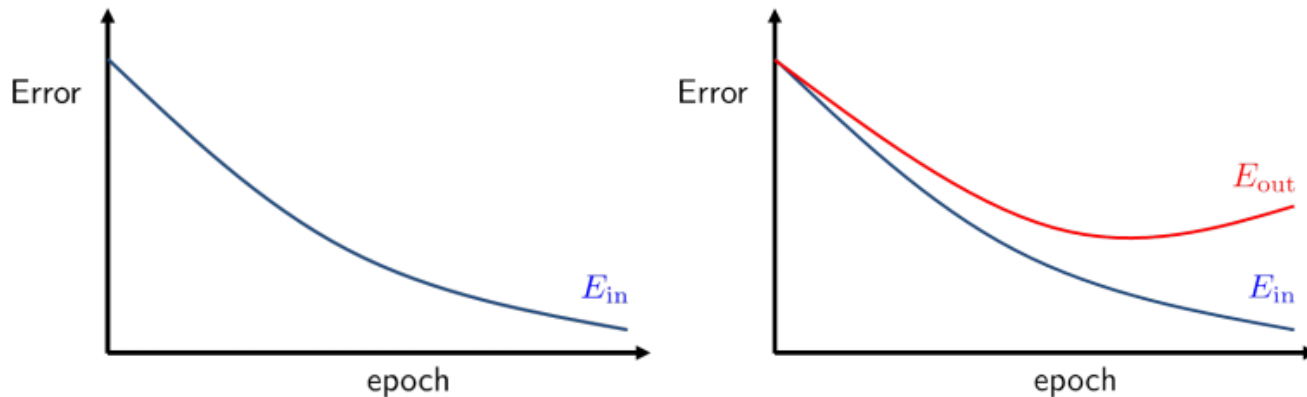
- If you use a **4-th order polynomial** to fit the data
- **$E_{in} = 0$** : You have a perfect fit
- **$E_{out} = \text{terrible}$** . You cannot generalize

What could go wrong?

Is Model Complexity the Reason?

Common belief: Complex models tends to overfit. Is this true?

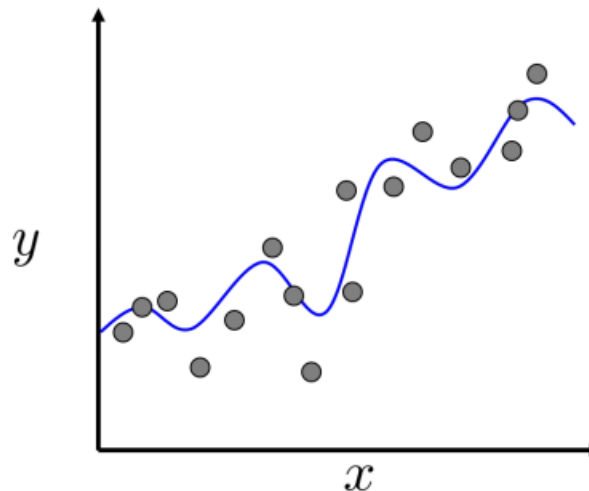
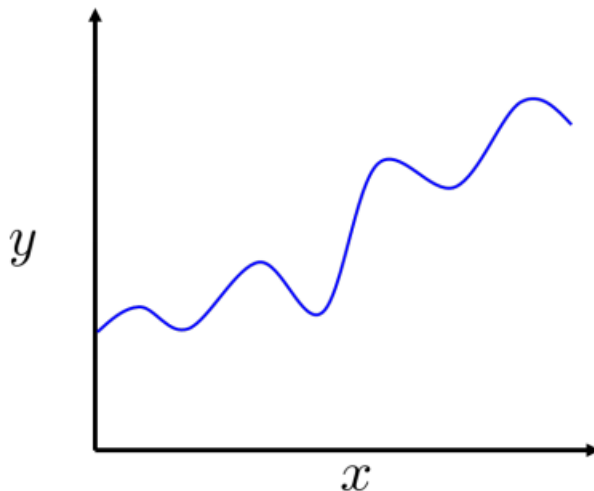
- Fix a neural network structure
- Fix training data
- Run for more epochs
- E_{in} drops
- E_{out} drops and then rises
- When you use more epoches, you fit the noise
- In this example, the network capacity is not changed, but you still have overfit



Is Noise the Reason?

Common belief: There is always noise, and so you overfit. Is it true?

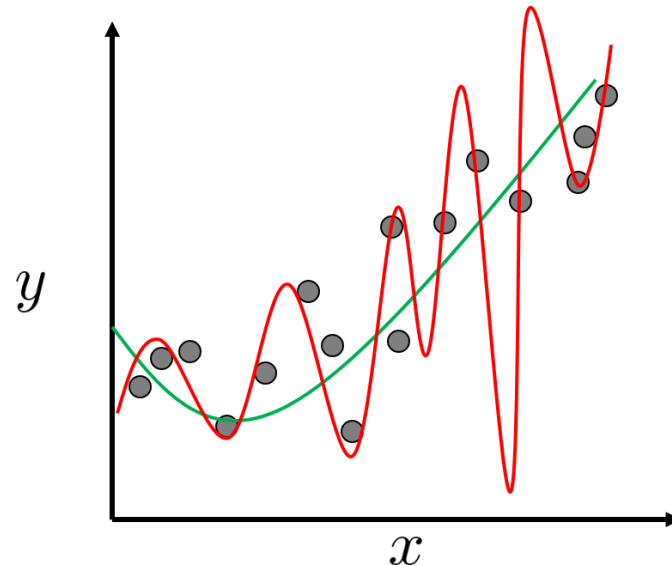
- You have a 10-th order target function
- You generate N noisy observations
- How well can you fit if you
 - (i) Use a 2nd order polynomial
 - (ii) Use a 10-th order polynomial



Is Noise the Reason?

Common belief: There is always noise, and so you overfit. Is it true?

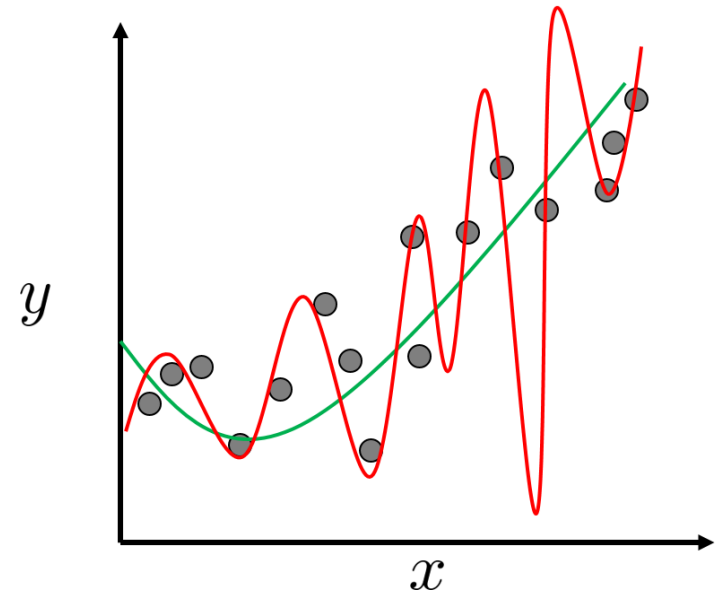
- You have a 10-th order target function
- You generate *N noisy observations*
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Is Noise the Reason?

- Let's look at **the training and testing error**.

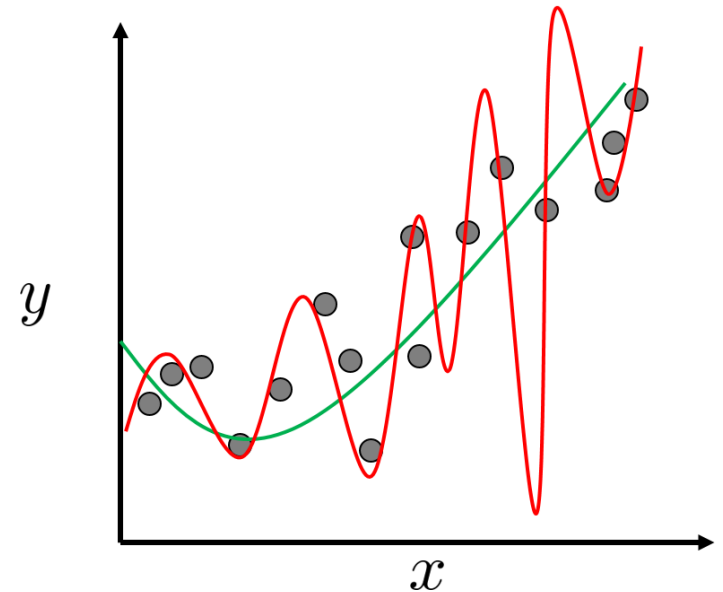
- You have a 10-th order target function
- You generate N noisy observations
- How well can you fit if you
 - (i) Use a 2nd order polynomial: $E_{in} = 0.05$
 - (ii) Use a 10-th order polynomial: $E_{in} = 0.034$



Is Noise the Reason?

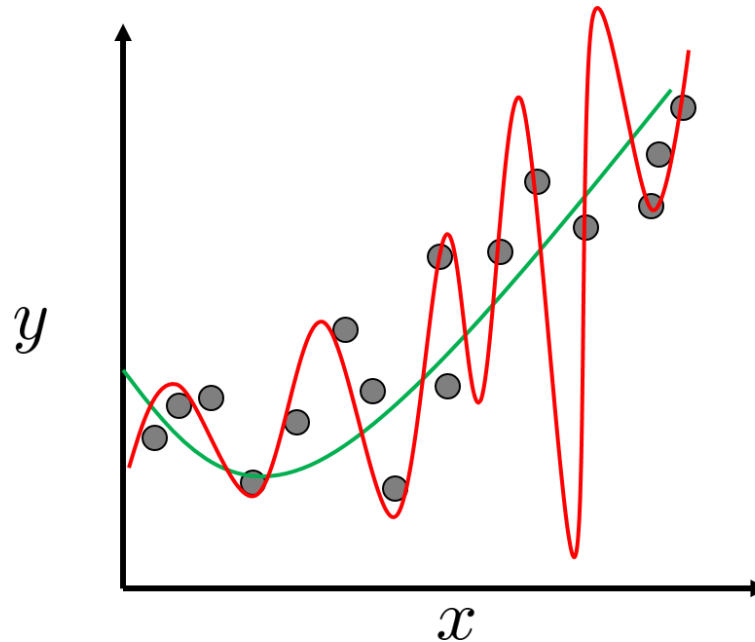
- Let's look at **the training and testing error**.

- You have a 10-th order target function
- You generate N noisy observations
- How well can you fit if you
 - (i) Use a 2nd order polynomial: **$E_{in} = 0.05$, $E_{out} = 0.127$**
 - (ii) Use a 10-th order polynomial: **$E_{in} = 0.034$, $E_{out} = 9.00$**



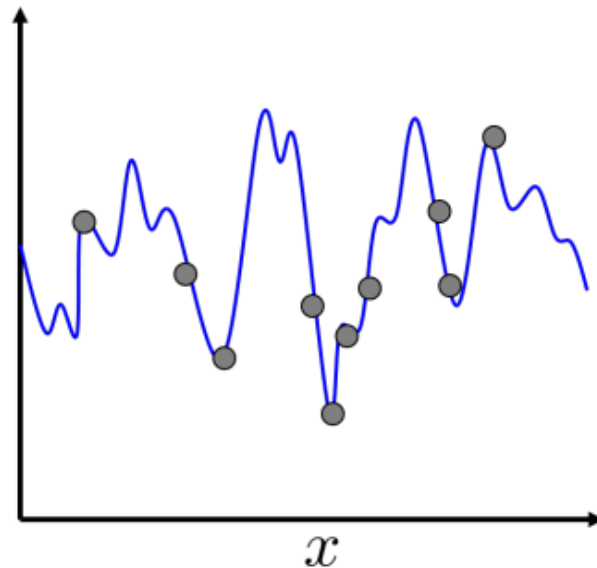
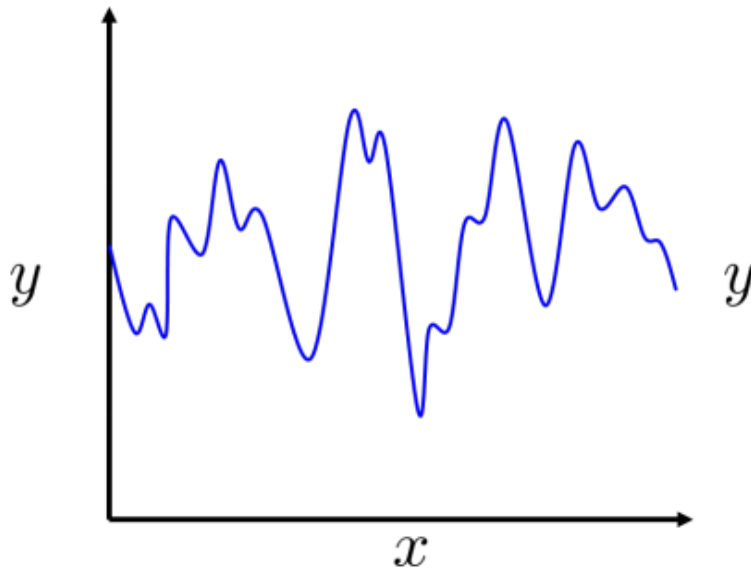
Is Noise the Reason?

- Noise is indeed a reason for overfitting
- You are fitting noise if you use complex models
- **Trade-off:** noise and model complexity
- But not the only reason



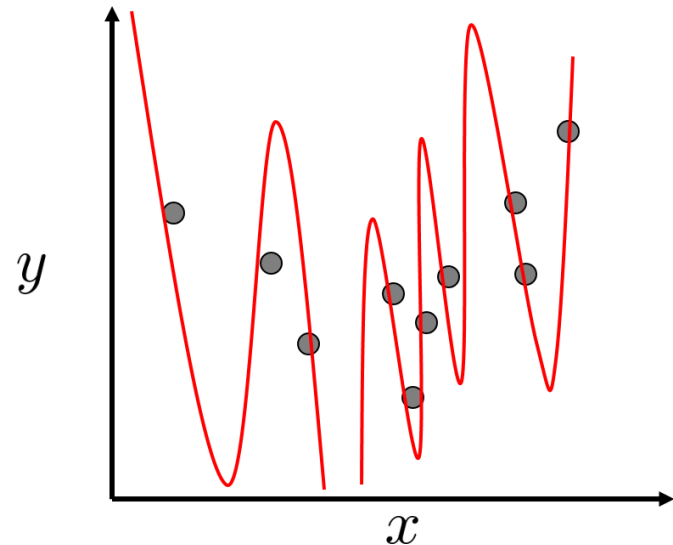
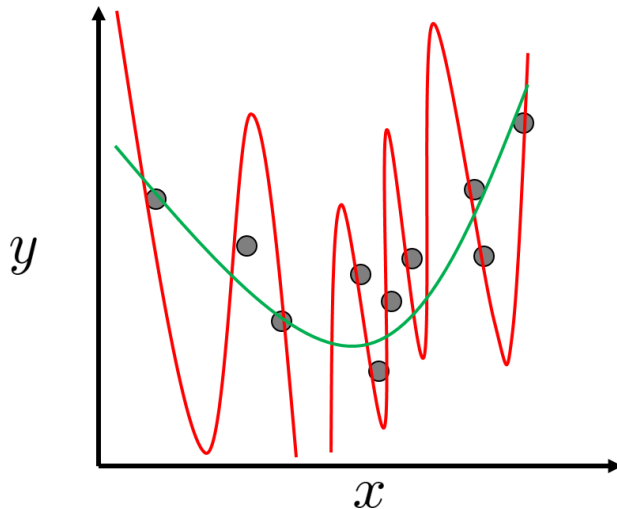
How about we Consider a Clean Target?

- You have a 50-th order target function
- You generate ***N clean observations***
- How well can you fit if you
 - (i) Use a 2nd order polynomial
 - (ii) Use a 10-th order polynomial



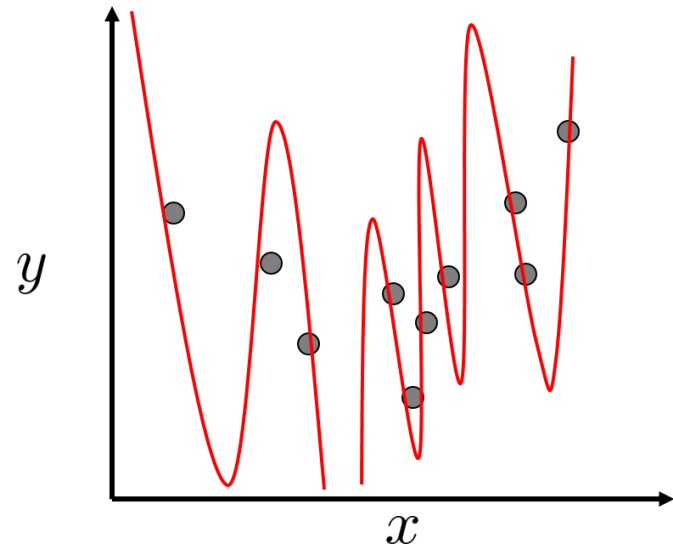
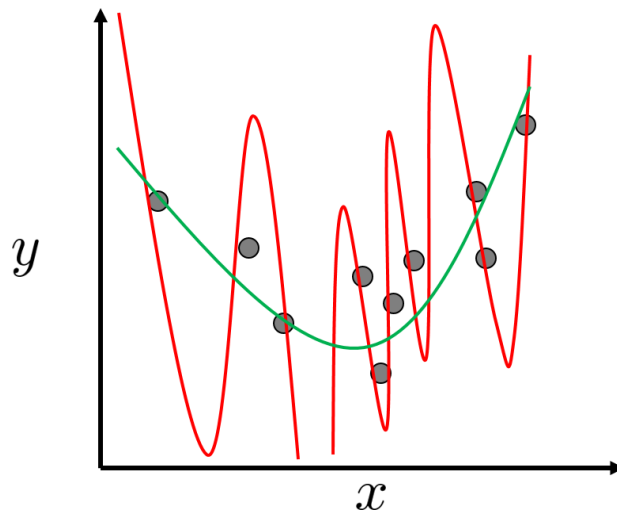
How about we Consider a Clean Target?

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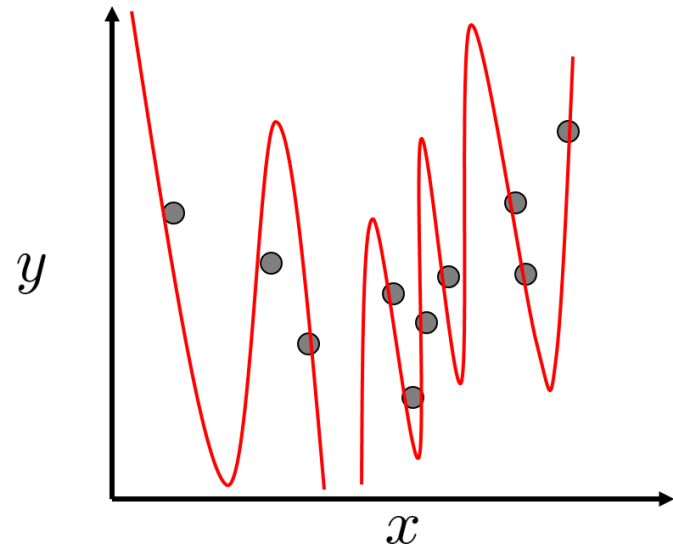
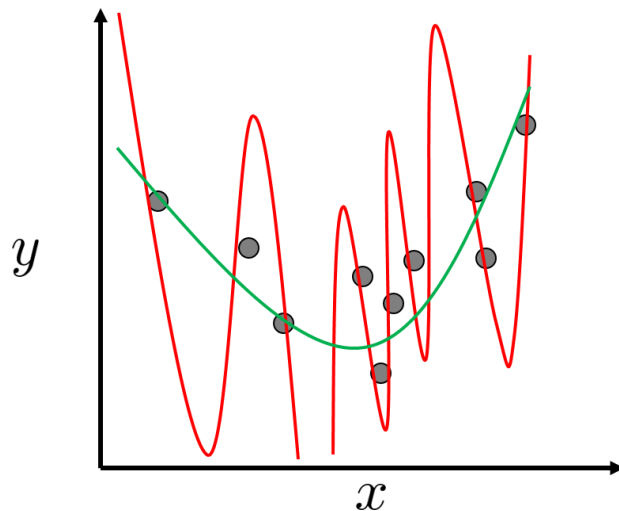
How about we Consider a Clean Target?

- You have a 50-th order target function
- You generate *N clean observations*
- How well can you fit if you
 - (i) Use a 2nd order polynomial: $E_{in} = 0.029$
 - (ii) Use a 10-th order polynomial: $E_{in} = 10^{-5}$

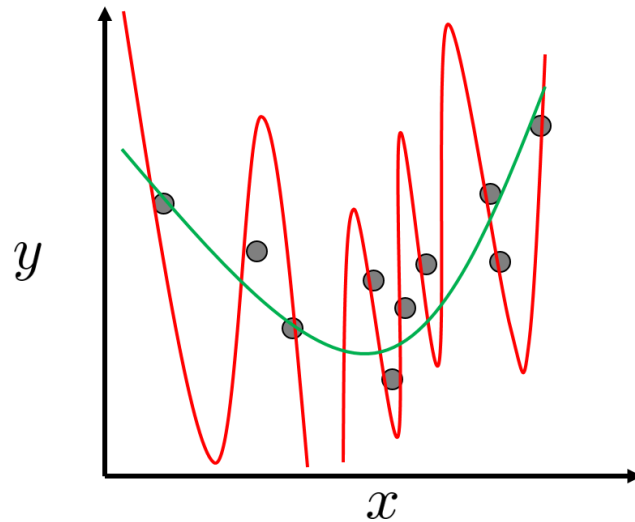


How about we Consider a Clean Target?

- You have a 50-th order target function
- You generate *N clean observations*
- How well can you fit if you
 - (i) Use a 2nd order polynomial: $E_{in} = 0.029$, $E_{out} = 0.120$
 - (ii) Use a 10-th order polynomial: $E_{in} = 10^{-5}$, $E_{out} = 7680$



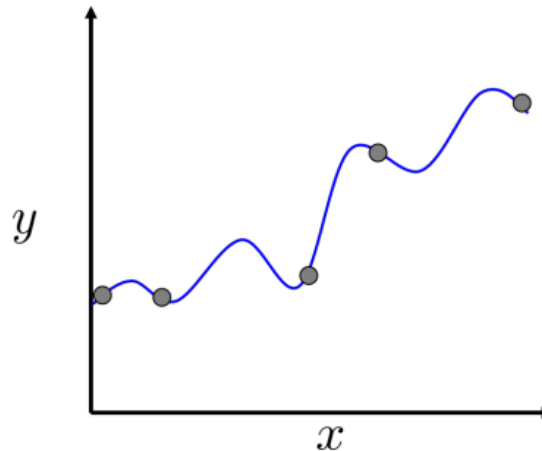
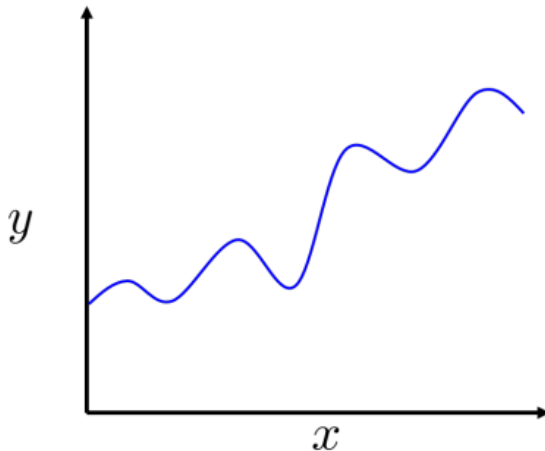
Is Noise the Reason?



- Noise-free
- Still overfit
- **Problem 1:** Model mismatch? 50th order target and 10th order fit?
- **Problem 2:** Lack of training samples?

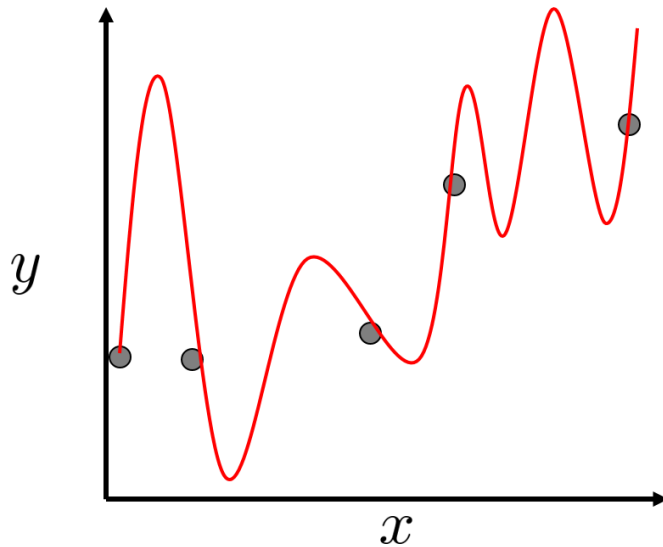
Is Model Fit a Reason?

- What if we know that the target function is 10-th order
- How well can you fit if you
 - (i) Use a 2nd order polynomial
 - (ii) Use a 10-th order polynomial
- The samples are clean

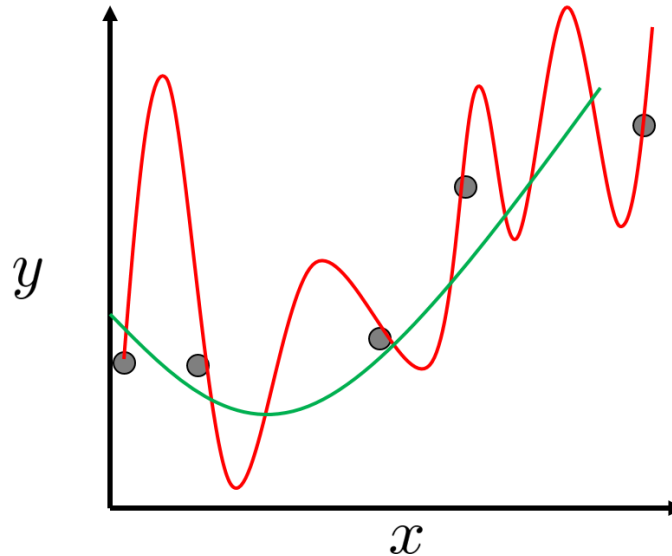


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Is Model Fit a Reason?

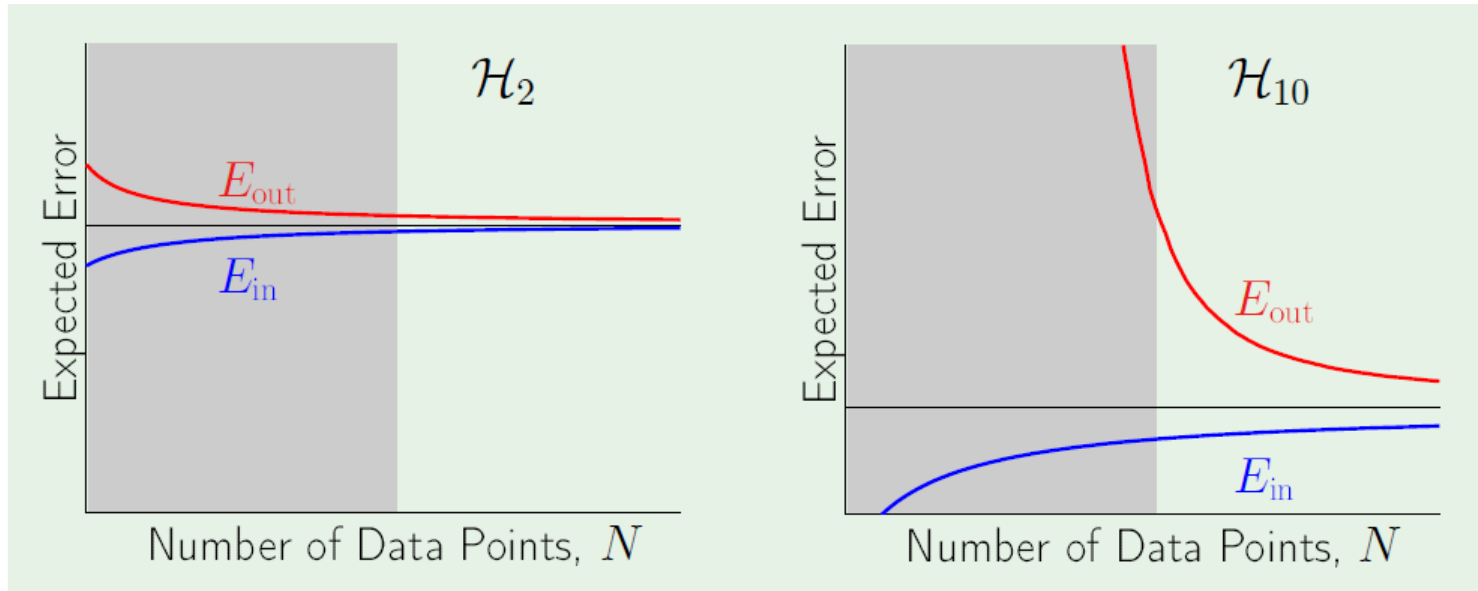


- Noise-free
- Model match
- Does not fit well
- **Problem:** Lack of training samples

What Causes Overfit?

- It is the trinity of
- **Training samples**
 - Not enough training samples?
 - Impossible to fit a complex model
 - Impossible to deal with noise
- **Target complexity**
 - Very complex target?
 - Need a lot of training samples to fit well
 - Not enough training samples, then can only use less complex hypotheses
- **Noise**
 - A lot of noise?
 - Need more training samples to average out the effect of noise

Learning Curve



Noise free

When N is small, $\mathcal{H}_2$ has lower E_{out}

$\mathcal{H}_2$ has higher steady state than $\mathcal{H}_{10}$

Bias-Variance

Recall this derivation:

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} \left[\mathbb{E}_{\text{out}}(g^{(\mathcal{D})}) \right] \\ = & \mathbb{E}_{\mathbf{x}} \left[\mathbb{E}_{\mathcal{D}} \left[g^{(\mathcal{D})}(\mathbf{x})^2 \right] - 2\mathbb{E}_{\mathcal{D}}[g^{(\mathcal{D})}(\mathbf{x})]f(\mathbf{x}) + f(\mathbf{x})^2 \right] \\ = & \mathbb{E}_{\mathbf{x}} \left[\underbrace{\mathbb{E}_{\mathcal{D}} \left[g^{(\mathcal{D})}(\mathbf{x})^2 \right] - \bar{g}(\mathbf{x})^2}_{\mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]} + \underbrace{\bar{g}(\mathbf{x})^2 - 2\mathbb{E}_{\mathcal{D}}[g^{(\mathcal{D})}(\mathbf{x})]f(\mathbf{x}) + f(\mathbf{x})^2}_{(\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2} \right]. \end{aligned}$$

The bias and variance are defined as

$$\begin{aligned} \text{bias}(\mathbf{x}) &\stackrel{\text{def}}{=} (\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2, \\ \text{var}(\mathbf{x}) &\stackrel{\text{def}}{=} \mathbb{E}_{\mathcal{D}}[(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}))^2]. \end{aligned}$$

What if $f(\mathbf{x}) \leftarrow f(\mathbf{x}) + \epsilon(\mathbf{x})$, where $\mathbb{E}[\epsilon(\mathbf{x})] = 0$?

Bias-Variance with Noise

We can show the following:

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}, \epsilon} \left[(g^{(D)}(\mathbf{x}) - (f(\mathbf{x}) + \epsilon(\mathbf{x})))^2 \right] \\ &= \mathbb{E}_{\mathcal{D}, \epsilon} \left[(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) + \bar{g}(\mathbf{x}) - f(\mathbf{x}) - \epsilon(\mathbf{x}))^2 \right] \\ &= \mathbb{E}_{\mathcal{D}, \epsilon} \left[\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 + (\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2 + (\epsilon(\mathbf{x}))^2 \right] \end{aligned}$$

Cross-terms involving $\mathbb{E}[\epsilon(\mathbf{x})]$ is zero

So

$$\begin{aligned} E_{\text{out}} = \mathbb{E}_{\mathbf{x}}[\odot] &= \mathbb{E}_{\mathcal{D}, \mathbf{x}} \left[\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right] + \mathbb{E}_{\mathbf{x}} \left[(\bar{g}(\mathbf{x}) - f(\mathbf{x}))^2 \right] \\ &\quad + \mathbb{E}_{\mathbf{x}, \epsilon} [\epsilon(\mathbf{x})^2] \end{aligned}$$

Bias-Variance with Noise

$$E_{\text{out}} = \mathbb{E}_{\mathcal{D}, \mathbf{x}} \left[\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right] + \mathbb{E}_{\mathbf{x}} \left[\left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right] + \mathbb{E}_{\mathbf{x}, \epsilon} \left[\epsilon(\mathbf{x})^2 \right]$$

Variance: $\mathbb{E}_{\mathcal{D}, \mathbf{x}} \left[\left(g^{(D)}(\mathbf{x}) - \bar{g}(\mathbf{x}) \right)^2 \right]$

Bias: $\mathbb{E}_{\mathbf{x}} \left[\left(\bar{g}(\mathbf{x}) - f(\mathbf{x}) \right)^2 \right]$

Noise: $\mathbb{E}_{\mathbf{x}, \epsilon} \left[\epsilon(\mathbf{x})^2 \right]$

Overfitting $\downarrow$ if number of data points $\uparrow$

Overfitting $\uparrow$ if noise $\uparrow$

Overfitting $\uparrow$ if target complexity $\uparrow$

Summary

Overfit happens because of noise, target complexity and training samples.

Overcoming overfit requires:

- Reduce the amount of noise in data (Could be hard)
- Reduce target complexity (May not be possible)
- Increase training samples

What else can we do?

- Choose a low complexity model even though target complexity is high
- Regularize the model complexity by promoting low order models

Overcoming Overfit



- Regularization is one weapon to combat overfitting.
- Constrains the learning algorithm to improve out-sample error when noise is present.
- Regularization is as much an art as it is a science.

Regularization from VC Analysis

$$E_{\text{out}}(h) \leq E_{\text{in}}(h) + \Omega(\mathcal{H}), \quad \text{for all } h \in \mathcal{H}$$

- Model complexity penalty $\Omega(H)$

If you want $E_{\text{out}}(h)$ to be small, better to make $\Omega(H)$ small

Roughly speaking, fit data using a “simple” h from H

So you are effectively minimizing
minimize

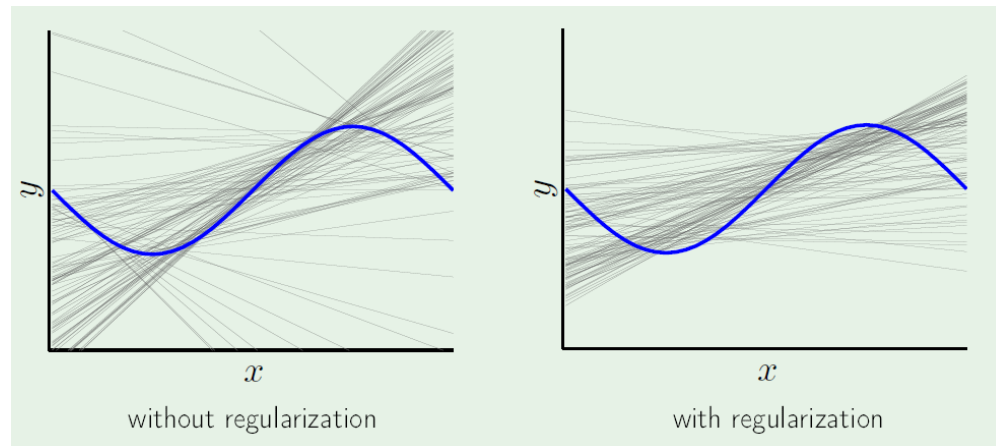
h

$E_{\text{in}}(h) + \Omega(h);$

That is, instead of minimizing $E_{\text{in}}(h)$ only, you minimize $\Omega(h)$ too

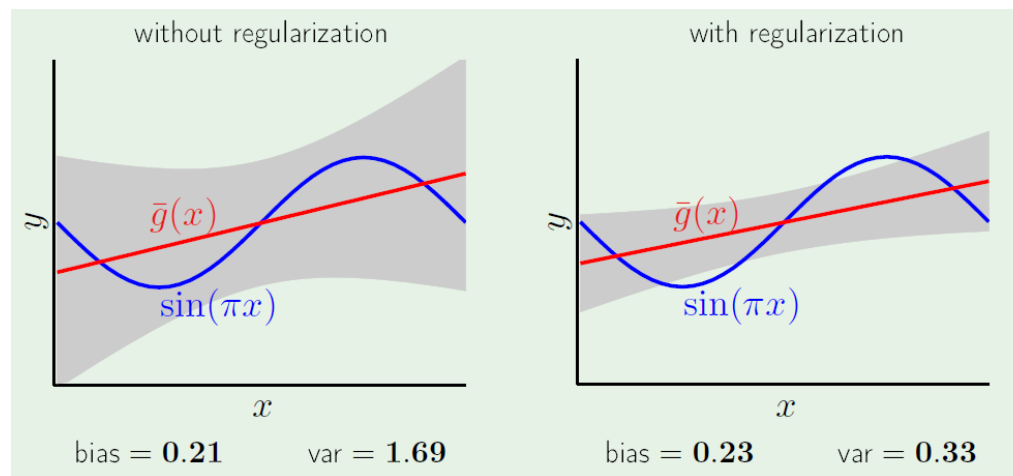
Example

- One regularization technique is weight decay.
- Measures the complexity of a hypothesis h by the size of the coefficients used to represent h (e.g., in a linear model).
- This technique prefers mild lines with small offset and slope.
- Applying this concept to the sine example before, trying to fit $N = 2$ data points, using H_1 (the set of lines)



Example

- Recall the constant model: Fit the two data points using a constant line.
- Constant model has $E_{out} = 0.75$
- Unregularized model has $E_{out} = 1.90$
- Regularized model has $E_{out} = 0.56$
- Bias–variance: Improve variance but suffer from bias. Overall is better.



Why Need Regularization?

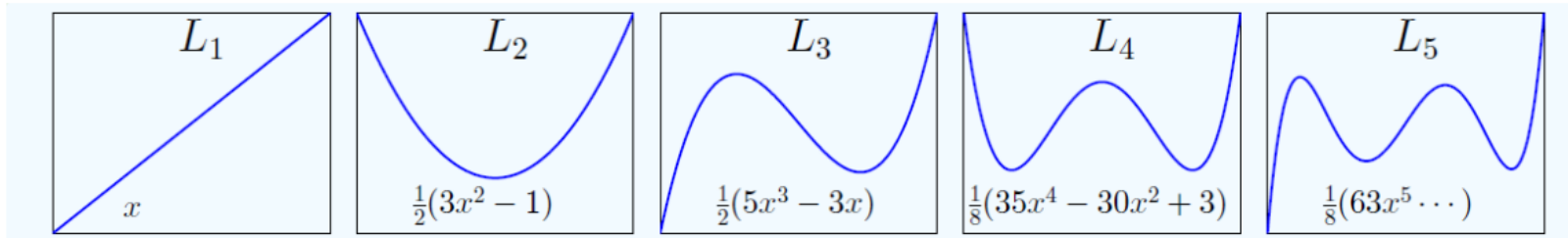
- The linear model is too sophisticated for the amount of data we have.
- A line can fit any two points!
- This problem is still here even if we change the target function.
- The need of regularization depends on quantity of data, and quality of data.
- Given only two points, we can either choose
 - a simple model, e.g., constant model
 - to constrain the model, e.g., weight decay
- Constraining the model gives us more flexibility

Soft Order Constraint

Consider the following example

- $\mathcal{H}$ = set of polynomials in one variable $x \in [-1, 1]$.
- E.g., $h(x) = 2x^2 + 3x + 7$.
- Want to express $h(x)$ using basis function.
- Basis functions for polynomials are Legendre polynomials $L_q(x)$, $q = 1, 2, \dots$
- So, any $h(x)$ can be expressed as

$$h(x) = \sum_{q=1}^Q w_q L_q(x) \quad (2)$$



Soft Order Constraint

This model is indeed linear! (Why?)

- You define a nonlinear transform Φ ,

$$\mathbf{z} = \Phi(x) = \begin{bmatrix} 1 \\ L_1(x) \\ \vdots \\ L_Q(x) \end{bmatrix}$$

- The hypothesis set is

$$\mathcal{H}_Q = \left\{ h \mid h(x) = \mathbf{w}^T \mathbf{z} = \sum_{q=0}^Q w_q L_q(x) \right\}$$

- So now you can define training error (for linear regression) as

$$E_{\text{in}}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (\mathbf{w}^T \mathbf{z}_n - y_n)^2$$

Soft Order Constraint

There are multiple ways of constraining the weights.

- **Hard** constraint:

- Force coefficients to be zero.
- For example,

$$\mathcal{H}_2 = \{\mathbf{w} \mid \mathbf{w} \in \mathcal{H}_{10}; w_q = 0, \text{ for } q \geq 3\}.$$

- **Soft** constraint:

- Force coefficients to be small.
- For example,

$$\sum_{q=0}^Q w_q^2 \leq C$$

- It encourages weights to be small without changing the order of the polynomial by explicitly forcing some weights to zero.